

Pool Adjacent Violator Algorithm (PAVA) for label position optimization

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1 Introduction

R packages like `directlabels` and `animint2` implement aligned text label drawing in a way that uses optimization to ensure readability. This paper describes the previous algorithm based on quadratic programming, and a new approach based on PAVA.

2 Problem

2.1 Data

We assume there are n labels. Each label $i \in \{1, \dots, n\}$ has two associated data: a target position $t_i \in \mathbb{R}$, and a half-size $h_i \in \mathbb{R}_+$. We assume the labels are already sorted by target value, $t_1 \leq \dots \leq t_n$.

Let $x_1, \dots, x_n$ be the label centers (optimization variables). For each label i , the lower limit of its text is $x_i - h_i$, and the upper limit is $x_i + h_i$. We assume the user inputs a global lower bound $\underline{B} \in \mathbb{R}$ and upper bound $\overline{B} \in \mathbb{R}$ in which text can be drawn.

2.2 Optimization

The label positions are determined as the solution of the optimization problem:

$$\begin{aligned} & \min_{x \in \mathbb{R}^n} ||x - t||^2 \\ & \text{subject to } x_1 - \underline{B} \geq h_1, \\ & \quad \forall i \in \{2, \dots, n\}, x_i - x_{i-1} \geq h_i + h_{i-1}, \\ & \quad \overline{B} - x_n \geq h_n. \end{aligned} \tag{1}$$

3 Solutions

3.1 Previous solution using quadratic programming

TODO.

3.2 Proposed solution using PAVA

The proposed solution involves using a change of variables and PAVA. First, let the lower limit vector be

$$L = \text{cumsum}(\underline{B} + h_1, h_1 + h_2, \dots, h_{n-1} + h_n). \quad (2)$$

Each element of the L vector represents the lower feasible limit of the corresponding element of x , given the constraints. In other words, $L \in \mathbb{R}^n$ is a vector, with elements defined as

$$L_i = \begin{cases} \underline{B} + h_1 & \text{for } i = 1 \\ L_{i-1} + h_{i-1} + h_i & \text{for } i \in \{2, \dots, n\}. \end{cases} \quad (3)$$

The change of variables is targets $d = t - L$, and solution $v = x - L$. In this new space, the feasible set is $v_i \in [0, U]$ for all i , where the upper limit is $U = \overline{B} - h_n - L_n$.

The corresponding optimization problem is

$$\begin{aligned} & \min_{v \in \mathbb{R}^n} \|v - d\|^2 \\ & \text{subject to } v_1 \geq 0, \\ & \quad \forall i \in \{2, \dots, n\}, v_{i-1} \leq v_i, \\ & \quad v_n \leq U. \end{aligned} \quad (4)$$

By using the definitions of d, v, L, U , it can be shown that (4) is equivalent to (1).

PAVA(d , lower, upper) is an algorithm that computes a solution to (4), given a vector of data d , and lower/upper bounds on feasible values for the solutions v . After computing the PAVA solution v , the solution to (1) is computed via $x = v + L$. Algorithm 1 summarizes the solution method.

Algorithm 1 PAVA for label position optimization

Require: half sizes $h_1, \dots, h_n \in \mathbb{R}$, target positions $t_1 \leq \dots \leq t_n \in \mathbb{R}$, bounds $\underline{B} < \overline{B} \in \mathbb{R}$.

$L \leftarrow \text{cumsum}(\underline{B} + h_1, h_1 + h_2, \dots, h_{n-1} + h_n)$

$U \leftarrow \overline{B} - h_n - L_n$

$v \leftarrow \text{PAVA}(t - L, 0, U)$

return $v + L$
